Inverse Problems and Imaging - 02624

# Homework 1

1. Consider the backwards heat equation modelled by the forward operator

$$Kx(t) = \int_0^\pi x(s)k(t,s)\mathrm{d}s \quad k(t,s) = \frac{2}{\pi}\sum_{n=1}^{\infty} e^{-Tn^2}\sin(nt)\sin(ns).$$

   (a) $K$ is compact in $L^2(0,\pi)$. Show that $K$ has the singular system given by

$$x_j(t) = \sqrt{\frac{2}{\pi}}\sin(jt), \quad y_j(s) = \sqrt{\frac{2}{\pi}}\sin(js), \quad \mu_j = e^{-Tj^2}.$$

   (b) Consider the approximation of $K$ by matrix $A \in \mathbb{R}^{n\times n}$ for $T = 0.03$ and $n = 128$. Compute the SVD of $A$. Compare in a plot the singular values of $K$ and $A$, and in a separate plot, the singular vectors for $A$ to the singular functions for $K$. Explain the differences.

   (c) Fix the function

$$x(s) = s\sin(s)\exp((s-\pi/2)^2).$$

   Plot (using the discrete approximations) the functions $x$ and $y = Kx$. Perturb $y$ by adding Gaussian noise $n$ of relative size $\|n\|_{L^2(0,\pi)}/\|y\|_{L^2(0,\pi)} = 2\%$ to obtain $y^\delta$, and make a Picard plot of the singular values $\mu_j$, the expansion coefficients $|(y^\delta, y_j)|$ and the reconstruction coefficients $|(y^\delta, y_j)|/\mu_j$. Use the Picard plot to decide for a truncation in a TSVD reconstruction; compute the reconstruction and compare the result to the true $x$.

2. Consider the Abel problem defined in $L^2(0,1)$ by

$$Kx(t) = \int_0^t \frac{1}{\sqrt{(t-s)}}x(s)\mathrm{d}s.$$

   With your a numerical discretization based on a quadrature rule, choose uniform $s-$ and $t-$meshes with $n = 512$ points to set up the matrix $A \in \mathbb{R}^n \times n, n = 512$, approximating $K$. Fix the function

$$x(s) = s^2.$$

al analysis - you can skip Show that $K$ is compact in $L^2(0,1)$.
   (Hint: Use the integral criterium $\int\int |k(t,s)|^2\mathrm{d}s\mathrm{d}t < \infty$, and create approximating sequence $K_\epsilon$ by of $K$ with $K_\epsilon$ compact.)

(a) Compute $y(t) = Kx(t)$ and discretize the function $y$ by a vector $y$ and add Gaussian noise of relative size $\delta = 5\%$ to obtain $y^\delta$

(b) Compute TSVD approximation $x_k$ for well chosen values in $k \in [1, 100]$ (or beyond) Create a plot of reconstructions, and compare it to the ground truth ($x$) via a plot of the relative error $(x - x_k)/\|x\|_2$. Explain why the TSVD solutions drop to zero at $s = 1$.

(c) Compute Tikhonov solutions $x_\alpha$ for well chosen regularization parameters $\alpha \in (10^{-4}, 10)$. Plot solutions and compare to TSVD.

(d) Try the above with a discontinuous function $x$.

3. Let $K$ be a compact from Hilbert space $X$ to Hilbert space $Y$ with singular system $\{x_j, y_j, \mu_j\}$. Assume that $K$ is injective. Let $\tilde{R}_\alpha$ be the operator in $H$ defined in the singular system for $K$ as

$$x_\alpha = \tilde{R}_\alpha y = \sum_j \frac{\mu_j^2}{\mu_j^3 + \alpha}(y, y_j)x_j.$$

(a) Show that $\lim_{\alpha\to 0} \tilde{R}_\alpha Kx = x$.
(Hint: use the expansion above to consider $\|\tilde{R}_\alpha Kx - x\|^2$ and make use of the Plancheral theorem. Take the limit $\alpha \to 0$; be careful when interchanging the limit and summation.)

(b) Consider the deblurring model problem for some $x$, matrix $A$, $y = Ax$, and $y^\delta = y + d$ with $\|d\| < \delta$. Choose the regularization parameter $\alpha(\delta) = \delta^3$. Demonstrate numerically (and if you like, theoretically,) that $\lim_{\delta\to 0} \|\tilde{R}_{\alpha(\delta)} y^\delta - x\| = 0$.

Important notes:

- Deadline for the HW1 is Friday October 24.
- All homeworks must be approved in order to qualify for the exam.
- Demonstrate in your solution your understanding of theory and computations.
- Each student should hand in a single pdf via Learn. The number of pages for the main report must not exceed 7. Code can be put in an appendix.
- Comment on all results: What do you see, and why is it reasonable (or unreasonable)?
- Make quality plots with proper names on axis, legends and captions. Zoom in on the relevant stuff.
- You might already now prepare the presentation of the HW to be used for the exam.